



Explaining variation in ammonia emission after field application of digested and untreated slurry using physical and chemical characteristics

Johanna Pedersen · Cristiane Romio · Andreas S. Pacholski ·
Sasha D. Hafner

Received: 16 April 2026 / Accepted: 1 September 2026
© The Author(s) 2026

Abstract Anaerobic digestion alters slurry properties with implications for ammonia (NH_3) emission after field application. While pH and dry matter (DM) are known drivers, they do not fully explain observed variability. This study quantified the extent to which chemical and physical characteristics explain NH_3 emission from digestates and untreated pig and cattle slurry. Five field trials were conducted over two years using nine digestates and six untreated slurries. Emission was measured over 150 h following surface band application using dynamic flux chambers and cavity ring-down spectroscopy. Slurries were characterized for pH, DM, rheological parameters (consistency coefficient K ; flow behavior index m), particle size distribution, and exposed surface area. Mixed-effects models were used to evaluate correlation between subsets of these variables and cumulative emission. Cumulative NH_3 emission was 41% of applied TAN on average for digestate, 27% for cattle, and 13% for pig slurry. Emission varied among trials

and was consistently positively correlated with pH. Similar positive associations were found for DM, viscosity (K or m), and the fraction of particles > 2 mm; however, these variables were collinear and provided comparable model support. Emission was negatively correlated with exposed surface area, likely reflecting reduced spreading and horizontal flow at higher DM, and showed limited predictive value. Overall, variation in NH_3 emission was largely attributable to pH and DM or correlated physical properties. Mechanistically, this reflects pH-dependent NH_3 speciation and reduced infiltration due to increased viscosity. Additional physical characterization beyond pH and DM offers limited predictive gain.

Keywords Anaerobic digestion · Ammonia volatilization · Biogas digestate · Liquid animal manure · Viscosity · Dynamic flux chamber

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10705-026-10524-x>.

J. Pedersen (✉) · C. Romio · S. D. Hafner
Department of Biological and Chemical Engineering,
Aarhus University, Blichers Alle 20, 8830 Tjele, Denmark
e-mail: jp@bce.au.dk

A. S. Pacholski
Thünen Institute for Climate-Smart Agriculture,
Brunswick, Germany

Introduction

Liquid and solid animal manure can be utilized for biogas production, a sustainable energy source when implemented at high technological level and using waste biomass as feedstock (Khoshnevisan et al. 2021; Lamolinara et al. 2022; Weiland 2009). After biogas production, the resulting digested slurry is typically applied to fields as an organic fertilizer (Lamolinara et al. 2022; Möller & Müller 2012; Nkoa 2013; Zilio et al. 2021).

To enhance biogas yields, slurry may be co-digested with a range of materials, including industrial waste streams and low-degradability agricultural residues such as straw, grass, and deep litter (Chuenchart et al. 2024; Møller et al. 2022). As the biogas sector continues to expand globally (Abanades et al. 2022), increasing attention is being given to sourcing feedstocks that do not compete for agricultural land. In many regions, land scarcity and competing demands for food production make it more sustainable to rely on co-products and residual biomass rather than crops grown specifically for biogas production.

The anaerobic digestion process alters slurry by lowering the dry matter (DM) concentration while increasing pH (Pedersen & Hafner 2023). A decrease in dry matter concentration reduces the ammonia emission potential when slurry or digestate is applied to the field (Häni et al. 2016; Misselbrook et al. 2005), presumably due to improved infiltration. However, an increase in pH raises the emission potential by increasing the concentration of the volatile free ammonia species (Sommer & Husted 1995a). Overall, these two effects have been found to approximately counteract each other when slurry is digested without co-substrates, that is, digested slurry and the corresponding undigested slurry tend to exhibit similar ammonia emission after field application on average (Pedersen & Hafner 2023).

If co-digestion of slurry with recalcitrant biomasses, such as straw and deep litter, results in higher dry matter concentrations than in undigested slurry, ammonia emission after field application is likely to increase (Pedersen & Hafner 2023). Romio et al. (2024) reported that changes in co-substrates led to a 52% increase in the dry matter concentration of Danish digestates and an average increase in digestate pH in Denmark from 7.87 to 8.04 during the period 2015–2023. This combination of higher dry matter and pH results in very high emission potentials (Pedersen & Hafner 2023), thereby increasing environmental risk.

In addition to dry matter and pH, several other factors affect ammonia emission following field application of digestate or slurry. These include weather conditions, soil properties, and application method (Hafner et al. 2025; van der Weerden et al. 2021). There are increasing international efforts to quantify differences both between and within different slurry categories in order to improve emission predictions

and inventories at farm and regional scales. However, data on digestate variables and associated emission remain limited, despite substantial and growing use of digestates as a fertilizer. Variation in key digestate variables, such as dry matter and pH, is considerable (Romio et al. 2024), making emission predictions uncertain, particularly for slurry-based digestates with high dry matter content.

The physical characteristics of slurry or digestate are crucial for the emission process, as they interact with soil properties and application methods and determine the infiltration of slurry or digestate into the soil and exposed surface area after application. Greater infiltration is generally associated with a lower emission (Häni et al. 2016; Misselbrook et al. 2005). The relationship between the exposed surface area and emission is more complex; however, all else being equal, a lower exposed surface area is assumed to reduce emission (Misselbrook et al. 2005; Webb et al. 2010). In most studies, dry matter concentration is the only measured physical characteristic of slurry or digestate (Bittman et al. 2005; Chantigny et al. 2009; Sanz-Cobena et al. 2019). Other parameters describing the physical characteristics and behavior of slurry and digestates are rarely included, and thus their importance as explanatory variables remains unknown. These parameters include apparent viscosity and particle size distribution. The viscosity of a liquid describes its resistance to flow and is defined as the ratio between shear stress (force per unit area) and shear rate (velocity gradient) (Schneider & Gerber 2020). It is hypothesized that viscosity and particle size distribution influence slurry and digestate infiltration into soil and the spreading of slurry and digestate on the soil surface.

Low-emission treatment and application techniques available for untreated slurry can also be applied to digestates (Lamolinara et al. 2022). Digestate can be injected at shallow or deep depths, which substantially reduces ammonia emission (Hafner et al. 2025; Webb et al. 2010); however, injection is not a viable option in growing cereal crops due to the risk of crop damage. Treatments that can effectively lower ammonia emission following surface application of digestates or slurry include acidification (Fangueiro et al. 2015) and solid–liquid separation (Pedersen et al. 2022). However, both techniques present challenges for digestates with high dry matter concentration. Effective pH reduction of digestates

typically require large amounts of acid (Nyord et al. 2021), increasing the risk of foaming, and complicating pumping and field application (J. Pedersen, personal observation). Furthermore, high acid usage is associated with a substantial cost. Solid–liquid separation can reduce emission, but separation efficiency and ammonia emission reductions are highly variable (Pedersen et al. 2022; Romio et al. 2024). In many cases, the dry matter concentration is not sufficiently reduced to achieve substantial ammonia emission reductions (Pedersen et al., 2026). The overall effect of separation on ammonia emission also depends on ammonia losses from the resulting solid fraction, which are often neglected when reporting separation effects.

For farmers receiving digestate as fertilizer, knowledge of product quality is crucial to ensure appropriate field management and to assess whether further treatment is warranted from an environmental or economic perspective (Epper et al. 2025). However, data on ammonia emission following field application of digestates remain limited, particularly for digestates with high dry matter concentration (Pedersen & Hafner 2023). The present study aimed to improve understanding of the relationship between digestate characteristics and ammonia emission following field application. A total of nine digestates were characterized using a comprehensive set of established but seldom measured physical and rheological parameters. Ammonia emission was measured from untreated slurry and digestate using a dynamic chamber system. The use of a small-plot dynamic flux chamber

measurement approach facilitated evaluation of the following hypotheses by supporting replication and sufficient precision for statistical comparisons between treatments.

This study hypothesized that seldom measured physical descriptors, such as digestate viscosity and particle size distribution, offer improved predictive power over, or in combination with, conventional predictors such as dry matter and pH in statistical models of ammonia emission. As these variables are linked to slurry–soil interaction processes (e.g. infiltration and surface residence time), they may better explain observed differences between slurry types, including between undigested slurries and digestates.

Materials and methods

A total of five field trials were conducted in 2023 and 2024 to quantify differences in ammonia emission after field application of digestate and pig and cattle slurry (Table 1). Digestates and slurries were applied by manual band spreading. Digestates from nine different biogas plants were used, along with cattle and pig slurry from three different commercial farms for each slurry type, giving a total of 15 different sources (i.e., plant or farm identity). All trials included digestate from the biogas plant at Campus Viborg, Aarhus University, Denmark (digestate A), which served as a reference. Ammonia emission was measured using a system of dynamic flux chambers (Pedersen et al. 2024a, b).

Table 1 Trial conditions. Soil water content ($\pm$ standard deviation ($n = 3$)), dry bulk density ($\pm$ standard deviation ($n = 3$)), soil pH ($n = 1$), ambient air temperature at application (based

on hourly averages) and average ambient air temperature during the measuring period with interval in parenthesis

Trial	Application start	Water content (g g ⁻¹)	Dry bulk density (g cm ⁻³)	Soil pH	Wheat height (cm)	Ambient air temperature (°C)	
						At application	Average over whole measuring period (min., max.)
1	2023-05-17 13:27	0.13 $\pm$ 0.03	1.17 $\pm$ 0.11	5.4	10	10.8	12.8 (4.1, 22.1)
2	2023-05-24 13:51	0.19 $\pm$ 0.01	1.03 $\pm$ 0.10	5.2	9	17.1	11.6 (5.9, 20.5)
3	2024-04-03 15:51	0.28 $\pm$ 0.02	1.18 $\pm$ 0.06	5.5	9	2.6	9.4 (1.1, 16.3)
4	2024-04-10 12:31	0.29 $\pm$ 0.01	1.15 $\pm$ 0.17	5.6	10	9.6	8.4 (1.9, 14.7)
5	2024-04-17 13:29	0.22 $\pm$ 0.01	1.11 $\pm$ 0.10	5.7	14	10.6	4.2 (− 1.7, 10.8)

Plot of air temperature in the whole measuring period can be found in Fig. S1

Chamber measurements modify the environment of the emitting surface compared with ambient conditions by altering air flow, precipitation, and solar radiation. Therefore, dynamic flux chamber measurements were used to assess relative differences between treatments rather than to estimate absolute in-situ emission (Hafner et al. 2024; Sommer & Misselbrook 2015).

Conditions during the trial

All the experiments were conducted in the same field at Campus Viborg, Aarhus University, Denmark (56.49316° N, 9.56128° E). The flat field had a loamy sand texture (9% clay, 24% silt, 65% sand), and winter wheat had been sown in the preceding autumn in both experimental years. The wheat was cut prior to all trials to minimize crop effects on emission (Table 1).

On the day of sample application, a top-soil sample was taken for measurement of 1:1 water pH, and three undisturbed soil cores (100 cm³) were taken from the topsoil layer (top 5 cm) to determine dry bulk density and gravimetric soil water content (measured by drying at 105 °C) (APHA 2017). Each trial required an area of approximately 20 m × 20 m, and soil homogeneity was assumed within this area. A new section of the field was used for each trial. Soil properties and wheat height are presented in Table 1.

Ambient air temperature at 2 m height was logged continuously at a nearby weather station (<0.7 km distance) at 1 h intervals (Table 1 and Fig. S1).

Digestate and slurry characteristics and their application

Digestates from a total of nine biogas plants were used in the trials. Animal slurry was the primary feedstock for all plants, although slurry type and addition of other substrates varied. Digestates A, B, E, F, G, H, and I originated from plants using straw as a co-substrate, with higher amounts in B, E, F, and G than in the others, although exact quantities were unknown. Digestate C and D did not include straw in the feedstock. Reactor temperatures were approximately 52 °C for all plants. Three cattle and three pig slurry samples were collected from six different commercial farms in Denmark. All digestates and slurries were collected 2–4 weeks prior to application and

stored outdoors under a roof in closed buckets until application.

The target total ammoniacal nitrogen (TAN) application rate was 60 kg ha⁻¹ for all digestates and slurries. The target volumetric application rates (m³ ha⁻¹) were calculated based on the TAN concentrations determined photometrically 1–2 days prior to application in a well-mixed sub-sample using a commercial kit (Spectroquant Test kit No. 1.00683.0001, Merck KGaA, Darmstadt, Germany). Subsamples were collected in the field for analysis. Actual TAN application rates, calculated from the applied volume and TAN concentration determined subsequently in laboratory analysis of field subsamples ranged between 44 and 73 kg TAN ha⁻¹ (Table S1).

In the laboratory, digestate and slurry samples were analysed gravimetrically for dry matter (DM %) and volatile solids (VS %) concentrations by heating at 105 °C and calcination at 550 °C (APHA 2017). TAN was measured as described above and total nitrogen was analysed using a low-field nuclear magnetic resonance sensor (Jensen et al. 2021). pH measurements (Knick Portavo 902 pH meter) were conducted in the laboratory (Table S1) and at the start of the application trials.

Apparent viscosity was measured in triplicate with a rotational viscometer (Brookfield DV2T, AMETEK Brookfield, Middleboro, USA). Approximately 600 mL of sample at 20 ± 3 °C was used. Cylindrical spindles (61 LV, 66 LV-C, 67 LV-C, and 64 LV) were selected according to the apparent viscosity range of each sample. Apparent viscosity was measured for one minute at different shear rates (proportional to the spindle rotational speed), increasing from 1 to 44 s⁻¹. As recommended by the manufacturer, viscosity values measured at torques lower than 10% were discarded.

The relationship between apparent viscosity (η in cP) and shear rate (γ in s⁻¹) was described by the power law model (Eq. 1) (Schneider & Gerber 2020):

$$\eta = K\gamma^{m-1} \quad (1)$$

Here, K is the consistency coefficient (in cP s^m) and m is the flow behaviour index (dimensionless). The consistency coefficient indicates the magnitude of viscosity, while the flow behaviour index describes deviation from Newtonian behaviour. A fluid is Newtonian when the apparent viscosity is independent

of shear rate ($m = 1$). Digestates are typically non-Newtonian and exhibit shear-thinning behaviour ($m < 1$), meaning viscosity decreases with increasing shear rate due to progressive particle alignment and reduced particle interactions (Schneider & Gerber 2020). Parameters K and m were estimated for each sample by minimizing the sum of squares difference between measured and predicted η values using Microsoft Excel's Solver.

Particle size distribution was quantified in duplicate by a combination of gravimetric and laser diffraction methods, since laser diffraction required particles < 2 mm. Approximately 30 g of sample was sieved through a 2 mm mesh and washed with approximately 500 ml deionized water. The retained material was dried at 105 °C for 24 h and weighed to determine the fraction of dry matter > 2 mm relative to total wet sample weight. The filtrate was analysed by laser diffraction (Malvern Mastersizer 2000, Malvern Panalytical, Malvern, United Kingdom) using the Fraunhofer scattering model after dispersion with water. From this, the percentage of particles < 1 mm among those < 2 mm was determined.

Digestate and slurry were applied manually to the soil surface with a watering can fitted with a hose to mimic trailing hose application. Previous comparisons have shown that variability in cumulative emission is lower with manual than machine application (Pedersen et al. 2024a); therefore, manual application was used to reduce replication requirements and allow inclusion of more treatments per trial. The predetermined volume of digestate or slurry was evenly distributed in two narrow bands within the area covered by each dynamic flux chamber.

Ammonia emission

Ammonia emission was quantified for 150 h following application using 16 dynamic flux chambers connected to a cavity ring-down spectrometer (CRDS) (model G2103, Picarro Inc., Santa Clara, CA, USA) to measure ammonia concentrations in inlet and outlet air streams. Detailed system description and validation are provided in Pedersen et al. (2024a). The CRDS instrument has been evaluated for potential interferences of other gases in an agricultural environment by Kamp et al. (2019) who found that any interferences were negligible.

Each chamber consisted of an open-bottom opaque polyethylene cylinder (700 mm diameter, 392 mm height). Air entered via three inlet pipes (1 m height, 80 mm diameter), positioned evenly on the top. A plywood deflector plate positioned 92 mm below the top ensured even airflow distribution (Figs. S2 and S3). Air was drawn from the chamber centre by a fan. An iris diaphragm controlled the volumetric air exchange rate (AER, min^{-1}), which was maintained at 15 min^{-1} throughout all trials.

Air was drawn from each chamber and from three background ports positioned at the air inlet of three of the chambers, at 1.5 L min^{-1} through a heated and insulated polyvinylidene fluoride (PVDF) tubing (6.35 mm OD, 4.76 mm ID) to the CRDS instrument. Each sample position was measured for 8 min per cycle. The CRDS measured the ammonia concentrations every 2 s. The mean ammonia concentration over the final 30 s of each cycle was used for flux calculations, as this period has been found to yield stable readings (Pedersen et al. 2024a).

Ammonia flux (F , $\text{mg N m}^{-2} \text{ h}^{-1}$) was calculated from the background-corrected ammonia concentration (C , mg N m^{-3}), airflow through the chamber (q , $\text{m}^3 \text{ h}^{-1}$), and the area of soil surface covered by the chamber (A , m^2) (Eq. 2).

$$F = \frac{C \cdot q}{A} \quad (2)$$

The relative flux (frac. applied TAN h^{-1}) was calculated by dividing flux ($\text{mg N m}^{-2} \text{ h}^{-1}$) by the TAN application rate (mg N m^{-2}). Cumulative emission was calculated using the trapezoidal rule, which approximates the area under the emission-versus-time curve by summing the areas of trapezoids formed between consecutive measurement points (Simmons 1996). To avoid bias, no detection limit was applied to the data, as recommended by Helsel (2010), therefore, some negative values occurred in the absence of significant emission.

Three replicates per treatment (digestate or slurry source) were used in each trial. Ammonia concentrations were recorded approximately every 2.5 h per chamber. During trial 1 instrument malfunction caused data loss between 48.1 h and 69.6 h after application. No gap filling was applied for this period the same trapezoidal approach was used in the period for calculation of cumulative emission.

Slurry exposed surface area measurements

Exposed surface area (ESA), defined as the percentage of soil covered by digestate or slurry, was determined using an imaging method. The dark chamber setup used is described in detail in Pedersen et al. (2021a) and the updated imaging method used to quantify the exposed surface area is described in detail in Pedersen et al. (2024b). Digestate or slurry was mixed with fluorescent dye (Acid Yellow 7, 20 g L⁻¹), applied manually as described above, and immediately covered with a dark chamber illuminated by ultraviolet light. A digital image was captured, and exposed surface area was quantified using image analysis. Threshold values are reported in Table S2.

Statistics

Linear mixed-effects models were used to assess and quantify differences among slurry types and correlation between chemical and physical slurry variables and emission. The unit of analysis was the field trial × slurry source combination. The response variable was total cumulative emission over 150 h or 8 h, taken as the mean of results from 3 field plots (dynamic chambers) for each slurry source in each trial. Slurry properties were based on the mean of 2 or 3 subsamples.

As slurry variables were observed rather than experimentally manipulated, and correlations among predictors were high in some cases, it was not possible to uniquely attribute variability to individual predictors. A primary subset of predictor variables was therefore selected for the modeling work, including known drivers of emission (slurry pH and dry matter concentration) and potential new explanatory variables (viscosity parameters and particle size distribution), while keeping the number of variables small enough to make hypothesis testing plausible and limit spurious results related to correlation among predictor variables.

The following predictor variables were initially considered in the linear models: trial and source of digestate (i.e., plant or farm identity) as random effects, and slurry type (digestate, cattle, or pig), exposed surface area, pH, dry matter, particles < 1 mm (% particles in fraction < 2 mm), particles > 2 mm (% wet sample), *m* (viscosity parameter), and *K* (viscosity parameter) as fixed effects. The trial variable was expected to cover application timing effects

due to intrinsically accounting for temperature and soil and crop conditions.

Simple bivariate correlation was used to select secondary variable subsets from the 10 that were considered (see Fig. S4 for scatterplots and Pearson's *r* values). Most of the variables were highly correlated with dry matter (the absolute value of Pearson's *r* was 0.69–0.90 for exposed surface area, log₁₀*K*, *m*, and particles > 2 mm vs. dry matter concentration), making it impossible to quantify unique relationships to emission in models that included dry matter and others. Instead, slurry type and the following combinations of fixed effects were considered:

- A. Slurry type,
- B. pH, dry matter, and particles < 1 mm,
- C. pH, exposed surface area, particles < 1 mm, and particles > 2 mm,
- D. pH, log₁₀(*K*), particles < 1 mm,
- E. pH, *m*, and particles < 1 mm, and
- F. pH, particles < 1 mm, and particles > 2 mm.

The slurry type model A was included both to quantify observed differences among slurry types and provide a reference for the evaluation of models B–F, based on quantitative predictors. The full models were simplified by removing predictors that were unimportant (*p* > 0.1 from likelihood ratio test) or impossible to estimate (the random slurry source effect was always given a variance of zero). Names of the resulting simplified models in the following sections match the letters given above for the starting subset. Additionally, the importance of residual slurry type differences in emission not related to physical and chemical properties was evaluated by comparing these models to an analogous set with slurry type included as a predictor (models B+ through F+). Names for models for 8 h total emission have an “8” suffix.

Linear mixed-effects models were fit with the lmer() function from the lme4 package (v1.1.37) (Bates et al. 2015) in R (R Core Team 2025). Hypothesis tests on relationships between individual predictor variables and emission were assessed using a likelihood ratio test (LRT) with the drop1() or anova() function. Models were compared using the Akaike information criterion (AIC), with differences < 2 considered trivial and 4–7 substantial (Anderson & Burnham 2004). Marginal and conditional R² values

Fig. 1 Ammonia emission rate (as a fraction of applied TAN per h) over the first 50 h after application of digestate, cattle- and pig slurry. Standard deviation displayed in bands ($n = 3$). See Fig. S5 for ammonia flux during the whole trial. Each biogas plant has an individual letter. Dig. = Digestate. Due to instrument malfunction, there is a data gap for trial 1

(Nakagawa & Schielzeth 2012) and root mean square error (RMSE) were also calculated for assessing model performance using the `model_performance()` function from the `performance` package (v0.16.0) in R (Lüdtke et al. 2021).

Results

Emission of ammonia was highest immediately after application of digestate or slurry, with large variations in flux rate and variation within measurements from the same digestate or slurry (Fig. 1 and Table 2). Diurnal emission patterns were observed for most digestates throughout the measuring period (150 h) (Fig. S5).

Total ammonia emission was strongly correlated with slurry type ($p < 0.001$ for overall term from likelihood ratio test for the mixed-effects model A). Emission was highest from digestate, with an overall estimate of 41% of applied TAN from model A (Table 3). There was evidence of correlation within slurry sources; emission from the same source was more similar across trials than emission from different sources, with an estimated standard deviation among sources (2.3% of applied TAN) similar to that among trials (2.8%) (Table 3).

Slurry characteristics together explained a large part of the observed variation in total 150 h emission within and between slurry types. Five models with a single physical parameter and pH were explored (Table 4). They all performed similarly, although clear evidence of a correlation with emission was lacking for the exposed surface area and $\log_{10}(K)$ models from the LRT tests (Table 4). The estimated pH effect was very similar for all models, around 0.2 per pH unit, (20% of applied TAN per pH unit). Most models showed the expected sign of the apparent relationship: total 150 h emission was positively correlated with dry matter, particles < 1 mm, and $\log_{10}(K)$, but emission was unexpectedly negatively correlated with exposed surface area. Most of these models estimated

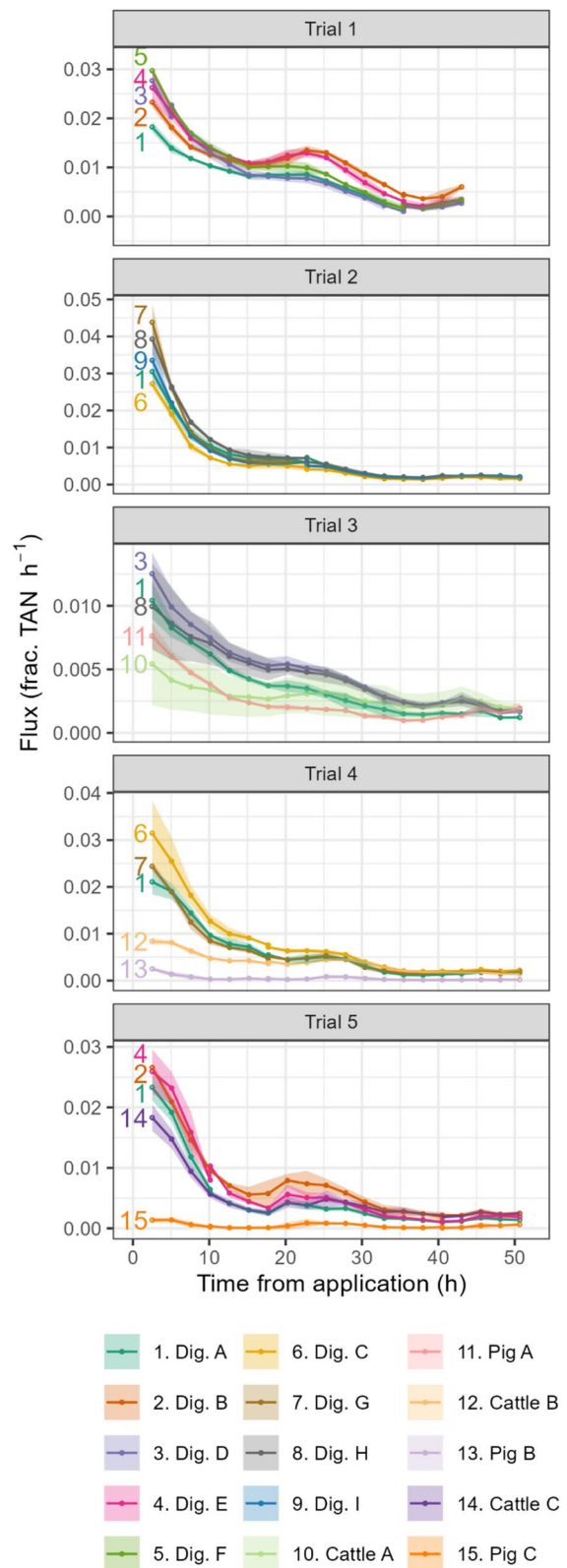


Table 2 Digestate and slurry variables (mean $\pm$ standard deviation ($n = 2$ for all except K and m where $n = 3$). For additional variables, see Table S1

Trial	Source ^a	Dry matter (%)	Particles > 2 mm (% wet sample)	Particles < 1 mm (% total particles in fraction < 2 mm)	K^b (cP s ^m)	m^c	Exposed surface area ^d (%)	pH ^e	NH ₃ emission (% TAN)
1	Digestate A	5.61 $\pm$ 0.05	0.96 $\pm$ 0.03	97.22 $\pm$ 0.03	853 $\pm$ 68	0.484 $\pm$ 0.006	16 $\pm$ 3	7.7 $\pm$ 0.1	36.6 $\pm$ 0.9
1	Digestate B	9.29 $\pm$ 0.0003	1.79 $\pm$ 0.09	93.98 $\pm$ 0.53	5966 $\pm$ 806	0.396 $\pm$ 0.022	13 $\pm$ 2	7.9 $\pm$ 0.1	53.1 $\pm$ 0.9
1	Digestate D	5.91 $\pm$ 0.04	0.33 $\pm$ 0.03	93.99 $\pm$ 0.75	617 $\pm$ 17	0.521 $\pm$ 0.018	11 $\pm$ 2	7.8 $\pm$ 0.03	41.4 $\pm$ 4.0
1	Digestate E	10.55 $\pm$ 0.03	2.81 $\pm$ 0.04	94.83 $\pm$ 0.20	1491 $\pm$ 231	0.346 $\pm$ 0.035	14 $\pm$ 1	7.7 $\pm$ 0.01	51.9 $\pm$ 3.2
1	Digestate F	6.48 $\pm$ 0.05	1.10 $\pm$ 0.04	97.42 $\pm$ 0.26	1062 $\pm$ 68	0.503 $\pm$ 0.020	14 $\pm$ 1	7.8 $\pm$ 0.01	49.3 $\pm$ 3.6
2	Digestate A	5.68 $\pm$ 0.01	0.76 $\pm$ 0.06	97.38 $\pm$ 0.48	931 $\pm$ 89	0.482 $\pm$ 0.026	18 $\pm$ 2	7.7 $\pm$ 0.02	41.1 $\pm$ 4.0
2	Digestate C	4.52 $\pm$ 0.003	0.29 $\pm$ 0.07	91.87 $\pm$ 0.28	181 $\pm$ 14	0.612 $\pm$ 0.022	17 $\pm$ 1	8.0 $\pm$ 0.02	35.2 $\pm$ 2.2
2	Digestate G	7.58 $\pm$ 0.02	1.15 $\pm$ 0.003	95.93 $\pm$ 0.47	552 $\pm$ 437	0.472 $\pm$ 0.012	14 $\pm$ 1	7.6 $\pm$ 0.03	47.1 $\pm$ 4.1
2	Digestate H	8.73 $\pm$ 0.24	0.73 $\pm$ 0.05	97.23 $\pm$ 0.13	446 $\pm$ 38	0.579 $\pm$ 0.045	13 $\pm$ 2	8.1 $\pm$ 0.06	49.4 $\pm$ 2.5
2	Digestate I	5.93 $\pm$ 0.003	0.66 $\pm$ 0.14	96.92 $\pm$ 0.76	370 $\pm$ 42	0.555 $\pm$ 0.038	18 $\pm$ 2	7.7 $\pm$ 0.1	43.2 $\pm$ 3.3
3	Digestate A	2.83 $\pm$ 0.01	0.11 $\pm$ 0.02	97.59 $\pm$ 0.66	6 $\pm$ 1	1.118 $\pm$ 0.039	63 $\pm$ 5	7.6 $\pm$ 0.1	27.3 $\pm$ 0.9
3	Digestate D	4.70 $\pm$ 0.02	0.05 $\pm$ 0.01	95.28 $\pm$ 0.45	54 $\pm$ 2	0.785 $\pm$ 0.009	38 $\pm$ 11	7.9 $\pm$ 0.01	34.0 $\pm$ 1.4
3	Digestate H	4.89 $\pm$ 0.05	0.23 $\pm$ 0.01	97.40 $\pm$ 0.16	55 $\pm$ 8	0.765 $\pm$ 0.030	55 $\pm$ 8	7.7 $\pm$ 0.04	33.4 $\pm$ 4.9
3	Cattle A	8.20 $\pm$ 0.13	1.36 $\pm$ 0.33	93.89 $\pm$ 1.06	2576 $\pm$ 69	0.474 $\pm$ 0.011	25 $\pm$ 5	7.0 $\pm$ 0.10	20.8 $\pm$ 8.5
3	Pig A	6.70 $\pm$ 0.21	0.27 $\pm$ 0.02	88.10 $\pm$ 0.01	403 $\pm$ 26	0.540 $\pm$ 0.022	26 $\pm$ 2	7.2 $\pm$ 0.01	24.3 $\pm$ 0.7
4	Digestate A	2.89 $\pm$ 0.003	0.03 $\pm$ 0.02	98.44 $\pm$ 0.19	17 $\pm$ 1	0.967 $\pm$ 0.024	46 $\pm$ 11	7.7 $\pm$ 0.01	37.4 $\pm$ 2.2
4	Digestate C	4.64 $\pm$ 0.05	0.22 $\pm$ 0.01	92.72 $\pm$ 1.34	318 $\pm$ 11	0.524 $\pm$ 0.019	37 $\pm$ 4	8.1 $\pm$ 0.02	49.3 $\pm$ 4.7
4	Digestate G	4.06 $\pm$ 0.002	0	99.88 $\pm$ 0.13	18 $\pm$ 0.4	0.977 $\pm$ 0.006	40 $\pm$ 3	7.8 $\pm$ 0.01	37.5 $\pm$ 2.0
4	Cattle B	10.40 $\pm$ 0.09	1.73 $\pm$ 0.33	94.96 $\pm$ 1.18	10,470 $\pm$ 2276	0.530 $\pm$ 0.358	11 $\pm$ 1	7.4 $\pm$ 0.3	23.5 $\pm$ 0.4
4	Pig B	2.70 $\pm$ 0.01	0.01 $\pm$ 0.00	96.50 $\pm$ 0.54	16 $\pm$ 0.5	0.972 $\pm$ 0.006	33 $\pm$ 2	7.0 $\pm$ 0.01	5.2 $\pm$ 1.9
5	Digestate A	2.43 $\pm$ 0.01	0.02 $\pm$ 0.01	98.06 $\pm$ 0.66	5 $\pm$ 1	1.222 $\pm$ 0.045	39 $\pm$ 5	7.5 $\pm$ 0.01	31.0 $\pm$ 1.5
5	Digestate B	7.72 $\pm$ 0.14	0.70 $\pm$ 0.01	96.05 $\pm$ 0.55	3044 $\pm$ 104	0.442 $\pm$ 0.014	16 $\pm$ 1	8.3 $\pm$ 0.01	42.9 $\pm$ 3.9
5	Digestate E	11.07 $\pm$ 0.15	3.53 $\pm$ 0.54	97.63 $\pm$ 1.00	6475 $\pm$ 806	0.148 $\pm$ 0.026	18 $\pm$ 2	7.7 $\pm$ 0.01	37.8 $\pm$ 3.9
5	Cattle C	7.87 $\pm$ 0.16	1.38 $\pm$ 0.35	95.24 $\pm$ 0.93	3921 $\pm$ 497	0.430 $\pm$ 0.016	20 $\pm$ 4	7.2 $\pm$ 0.3	31.9 $\pm$ 3.6
5	Pig C	3.57 $\pm$ 0.02	0.03 $\pm$ 0.01	95.42 $\pm$ 0.91	17 $\pm$ 0.2	0.965 $\pm$ 0.005	27 $\pm$ 2	6.2 $\pm$ 0.01	5.3 $\pm$ 1.3

^aBiogas plant, each biogas plant has an individual letter. ^bConsistency coefficient. ^cFlow behaviour index. ^dExposed surface area.^eMeasured in the laboratory

a similar variance for the random trial effect as was seen in the slurry type model (A) (Table 3), but interestingly none showed evidence of correlation within specific slurry sources after accounting for the correlation between predictor variables and emission as in the slurry type model A. All models based on slurry characteristics showed a better fit than the slurry type-only model A, with AIC values lower by at least 6 units (Tables 3 and 4). Addition of slurry type to the best fit slurry property models tended to improve models ($p = 0.011$ from an LRT for addition of slurry type for model C, $p < 0.01$ for models B, D, and E,

but $p = 0.12$ for model F (Table S3)). The AIC difference showed considerable evidence of an improvement for models B (difference of 5.8) and E (difference of 6.6), with small improvements in RMSE and marginal R^2 (Table S3). The model coefficient for pH changed substantially when slurry type was included: from 0.22 to 0.06–0.14 per pH unit (Table S3). With initial emission as the response variable (total over the first 8 h), correlation with physical predictors was not apparent; emission was clearly correlated with pH alone (coefficient 0.091 per pH unit, $p < 0.001$).

Table 3 Mixed-effects model parameter estimates and model performance for the slurry type model A

Component	Term	Value and std. err
Fixed effect	Digestate	0.41 (0.020)
	Cattle	0.26 (0.042)
	Pig	0.13 (0.045)
LRT <i>p</i> -value	Slurry type	< 0.001
Random effects (std. dev.)	Trial (intercept)	0.028
	Slurry source (intercept)	0.023
	Residual	0.063
AIC		−49.0
Conditional R ²		0.74
Marginal R ²		0.66
RMSE		0.056

The response variable was total cumulative emission (150 h after application) as a fraction of applied TAN (0.0–1.0) and the unit of analysis was a field trial × slurry source combination

Table 4 Mixed-effects model parameter estimates, model performance, and related information for physical and chemical parameter models B through F

Component		Model B	Model C	Model D	Model E	Model F
Fixed effects	Intercept	−1.35	−1.37	−1.34	−1.19	−1.35
	Physical parameter and units	Dry matter (%)	Exposed surface area (%)	$\log_{10}(K)$	<i>m</i>	Particles > 2 mm (%)
	Correlation with dry matter	–	−0.69	0.90	−0.85	0.88
	Physical value std. err	0.010 (0.0051)	−0.0022 (0.00096)	0.021 (0.014)	−0.11 (0.052)	0.030 (0.015)
LRT <i>p</i> -values	pH std. err	0.22 (0.031)	0.23 (0.032)	0.22 (0.032)	0.21 (0.031)	0.22 (0.030)
	Physical parameter	0.043	0.078	0.11	0.035	0.038
	pH	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Random effects (std. dev.)	Trial (intercept)	0.033	0.000	0.032	0.025	0.031
	Residual	0.061	0.068	0.063	0.062	0.061
AIC		−57.7	−56.7	−56.2	−58.0	−57.9
Conditional R ²		0.77	–	0.75	0.76	0.77
Marginal R ²		0.72	0.72	0.71	0.75	0.73
RMSE		0.055	0.064	0.058	0.058	0.056

Each includes slurry or digestate pH and a single physical variable. The response variable was total cumulative emission (150 h after application) as a fraction of applied TAN (0.0–1.0) and the unit of analysis was a single field plot

Visual evidence from plotted data supports results from the models. Positive correlation of cumulative ammonia emission with slurry dry matter concentration and pH was obvious, both between and within individual trials (Fig. 2). Results are similar for other physical variables (Fig. 3). Predictions from the dry matter model B were usually closer to measured emission values than the slurry type model A (Fig. 3 and S6). Based on the models, differences in pH, which account for the higher emission from digestate than the raw slurries,

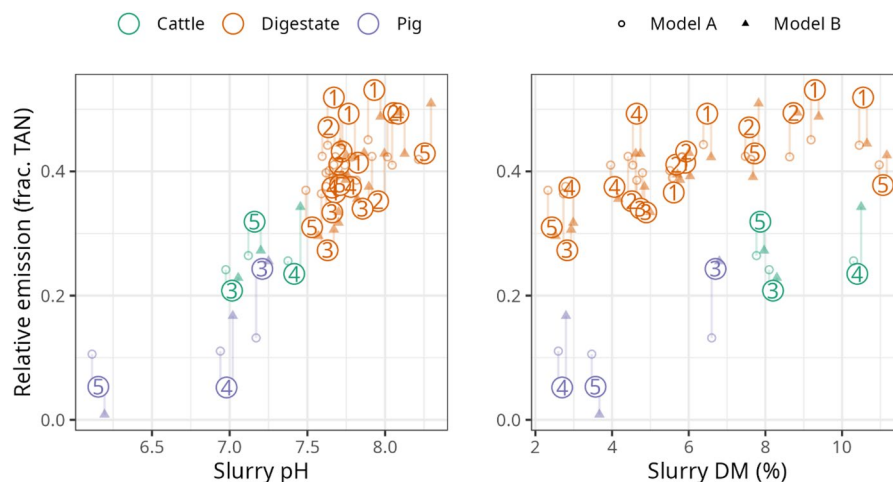
and differences in dry matter concentration explain much of the variability between different digestates.

Discussion

Digestate and slurry variables

The average dry matter concentration of the digestates of 6.1% (Table 2) was higher than the average

Fig. 2 Scatterplots of relative 150 h cumulative emission versus slurry pH (left) or dry matter (right) along with predicted values from models A and B. Integer values printed in circles show trial key. Colors show slurry types. Comparison of model-calculated and measured emissions for all models can be found in Fig. S6



value of 3.9% for published data according to a recent review (Pedersen & Hafner 2023). This likely reflects the increased use of high-dry matter biomasses at Danish biogas plants in recent years (Møller et al. 2022; Romio et al. 2024). The dry matter concentration of the pig slurries were close to the Danish average of 6.5%, whereas two cattle slurries (Cattle A and Cattle B) exceeded it (Adamsen & Hafner 2021) (Table 2).

The average pH of the digestates in the present study (8.0) was similar to literature values (8.0; Pedersen & Hafner 2023), the Danish average (7.9; Adamsen & Hafner 2021), and a recent Danish study (8.0; Romio et al. 2024). The pH of the cattle and pig slurries was likewise close to Danish averages (7.0 and 7.2, respectively; Adamsen & Hafner 2021). Overall, the slurries (including digestates) covered a wide pH range (Table 2), enabling assessment of pH effects on ammonia emission.

Consistent with Schneider and Gerber (2020) and Kariyama et al. (2018), digestate and slurry viscosity parameters varied substantially (Table 2). Viscosity parameters for digestates covered the full range observed for pig and cattle slurries (Figs. 3 and S4).

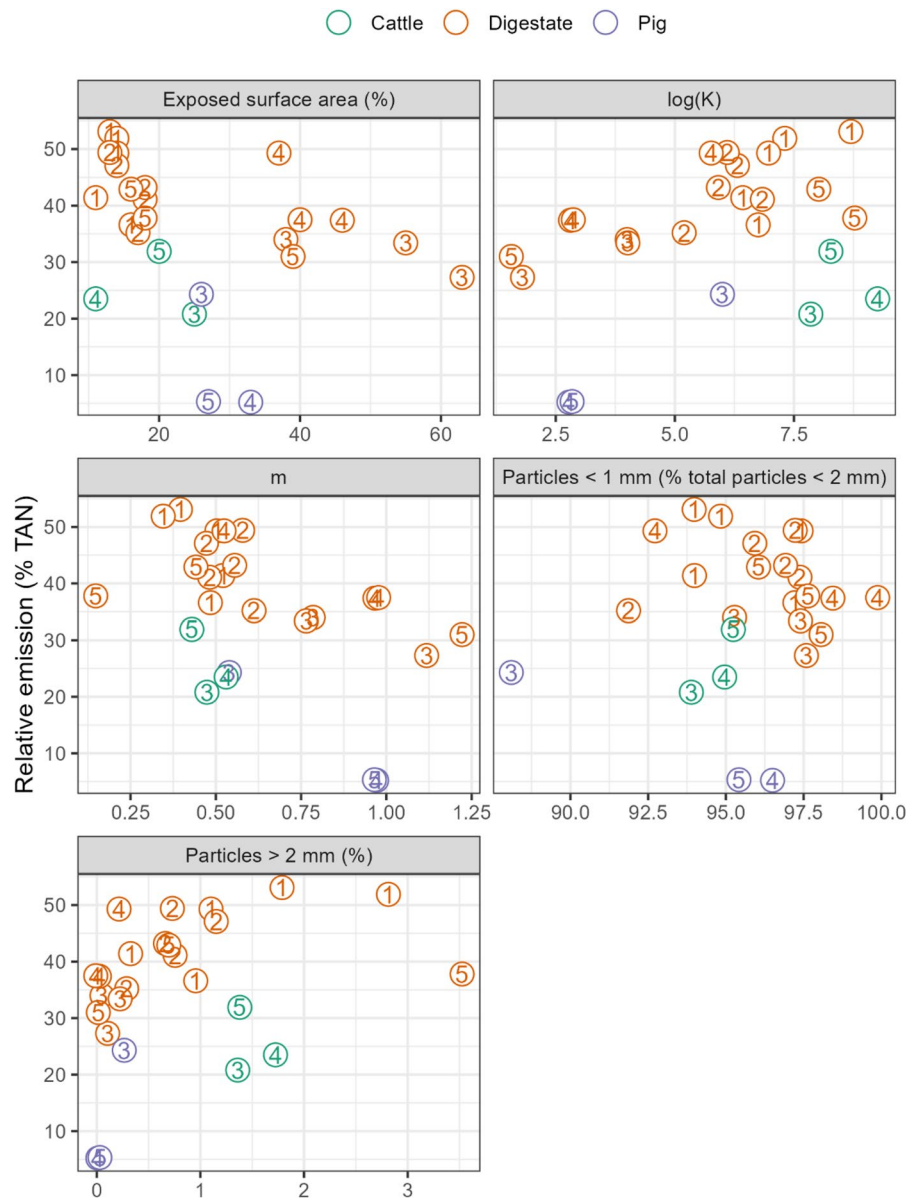
Cattle slurry exhibited high K (viscosity coefficient) and low m (non-Newtonian liquid, low flow behaviour index) values, whereas most pig slurry samples showed the opposite pattern, possibly reflecting lower dry matter concentrations. The viscosity parameter K generally increases with dry matter concentration (Fig. S4), consistent with increased particle resistance to flow (Langley-Randall et al. 2024). However, substantial variability in K persisted at high

dry matter concentrations (Table 2), indicating that dry matter alone does not fully explain flow behavior and that additional factors, such as particle size distribution and inter-particle interactions also play a significant role. The parameter m decreased with increasing dry matter (Table 2 and Fig. S4), indicating stronger shear-thinning behavior and that higher dry matter was associated with less Newtonian behavior. At low shear rates, particles have higher flow resistance until alignment reduces resistance at higher shear rates (Liu et al. 2019). Samples with low dry matter showed less pronounced shear thinning behavior and higher m values, reflecting more Newtonian behavior, implying that their viscosity is less sensitive to shear rate and therefore more uniform under varying flow conditions.

Predictors of ammonia emission

Analysis of emission and slurry variables strongly suggest that slurry pH and physical variables exert a major influence on ammonia emission (Table 4, Figs. 2 and 3). The pH effect is unambiguous, present as a clear positive correlation between total 150 h emission and pH in all models where it was assessed. This is consistent with the equilibrium shift towards free ammonia at higher pH (Sommer & Husted 1995a, 1995b). The effect of physical variables was less distinct. Previous field (e.g., Andersson et al. 2023; Chantigny et al. 2009; Frost et al. 1990) and laboratory (e.g., Sommer et al. 2006; Thompson et al. 1990) studies as well as analysis of compiled data (Hafner et al., 2018) have shown that

Fig. 3 Scatterplots of relative 150 h cumulative emission versus exposed surface area (%), $\log_{10}(K)$ (viscosity consistency coefficient), m (viscosity flow behaviour index), particles < 1 mm (% total particles < 2 mm), and particles > 2 mm (%). Integer values printed in circles show trial key. Colors show slurry types



emission generally increases with slurry dry matter concentration. Although this study aimed to clarify mechanisms and identify predictive physical variables, correlations among measured variables limited identification of independent effects. This suggests that inclusion of these variables in models will not provide any substantial accuracy advantage over dry matter determination, thereby rejecting the hypothesis of the study. The three alternative models including dry matter, intrinsic viscosity, or particles > 2 mm showed similar support ($\Delta AIC \approx 0$), indicating

that they capture a common underlying mechanism. Specifically, all three variables are proxies for slurry physical structure and solids content, which regulate infiltration dynamics: increasing dry matter, viscosity, or large particles fraction reduces slurry and TAN infiltration into the soil if all else is equal, thereby prolonging surface residence time and enhancing ammonia volatilization. The positive model coefficients for dry matter, particle concentration, and consistency coefficient and the negative coefficient for the flow behavior index are consistent with this

mechanism. While the results do not allow formal identification of a single dominant predictor among these correlated variables, dry matter effectively integrates these properties (Fig. S4). Given the simplicity of dry matter determination (mass loss after drying), it remains a practical and appropriate predictor for ammonia emission. Infiltration, however, depends on the interaction between slurry and soil properties, and all trials in the present study were conducted on a single soil type. The relative importance of dry matter and other physical slurry characteristics for infiltration, and hence for ammonia emission, may therefore differ on soils with contrasting texture, structure, or moisture content. Furthermore, the correlation between physical characters and emission breaks down for 8 h emission, suggesting that the physical properties are not as important for the initial emission rate, which is consistent with an effect mediated through infiltration. In contrast, a pH effect was clear even for 8 h emission.

Exposed surface area

Unexpectedly, emission was negatively correlated with exposed surface area (Table 2 and Fig. 3). A likely explanation is that high dry matter reduced both surface area (Fig. S4) and soil infiltration, increasing the volume of slurry remaining on the soil surface. Ammonia at the surface may be more susceptible to volatilization than ammonia infiltrated into soil, where mass transfer may be impeded or ammonia retained (Webb et al. 2010). This mechanism may outweigh effects of exposed slurry–air interface area. In some studies, a lower exposed surface area has been correlated to a lower cumulative emission (McCollough et al. 2022; Misselbrook et al. 2002; Smith et al. 2000). However, other studies or trials did not find this relationship (Misselbrook et al. 2005; Pedersen et al. 2021a, b; Salazar et al. 2014). A further complication arises from the surface area measurement method. The present method quantified a two-dimensional projection of slurry at the surface; actual exposed surface area may vary with microtopography despite identical projected two-dimensional area. The findings in the present study underline that the correlation between exposed surface area as projected to a two-dimensional soil surface and cumulative emission is not straightforward. With an unexpected negative correlation and less predictive

power than dry matter, exposed surface area is a poor predictor of ammonia emission after application.

Digestate versus untreated slurry

Emission differences between digestate and untreated slurry likely reflect the previously described offsetting effects of anaerobic digestion on dry matter and pH, together with the elevated dry matter of some digestates relative to the undigested slurries used here (Pedersen & Hafner 2023). In the current study, emission was highest from digestate and lowest from pig slurry (Table 3). The clear evidence that pH and dry matter (or related physical variables) affect emission from the present work and previous studies suggest that all emission differences between digestate and undigested slurry can be understood and even predicted based on slurry variables. This framework is partially consistent with the ALFAM2 model (Hafner et al. 2025) which includes pH and dry matter concentration effects but also lower emission from pig slurry compared to all other slurry types, such as cattle slurry or digestate. In contrast, in the present work it was digestate that was different from the other slurry types, and there was no evidence of a difference between cattle and pig slurry after adjusting for pH and dry matter concentration. Possible reasons for the discrepancy include nonlinear responses (the ALFAM2 model is nonlinear and the true relationships between emission and pH or dry matter almost certainly are as well). But the large change in the apparent pH response when slurry type was included in the models is indicative of confounding between slurry variables and types. This implies substantial uncertainty in statistical model parameter estimates when slurry type is included with pH and dry matter as predictors; the digestate difference is likely spurious. Beyond these average differences between slurry types, additional variability arose among individual trials and digestate sources, discussed further below.

Sources of variability

Besides digestate A, included in all the trials, additional biogas plant sources supplied slurry for trials in both years. Digestate variables varied substantially within source (Table 2), influencing emission differences. As an example, digestate C had similar dry matter concentration in trials 2 and 4, whereas

digestates A and G had lower dry matter in trial 4 than trial 2, accompanied by lower emission (Table 2). This highlights the importance of considering the variables of the digestate at the time of application, instead of assuming average values over long periods for the same digestate source. These sources of variability are shown clearly in the slurry type model A (Table 3); emission was related to both trial and source (i.e., biogas plant or farm identity), but the residual term standard deviation was nearly twice as large as those effects. The trial contribution may be related to soil conditions at the time of application, such as water content, as well as crop density. Furthermore, varying temperatures would inevitably also contribute to variation.

Diurnal emission patterns

Diurnal emission patterns were observed for digestates and cattle slurry, and for pig slurry with high dry matter in trial 3, even as late as seven days after slurry application (Figs. 1 and S5). It is proposed that the persistent diurnal pattern and emission observed after the initial days was due to the high dry matter concentrations of most of the digestates and cattle slurry, leaving more digestate or slurry at the soil surface to be subjected to temperature oscillations influencing emission. This is consistent with previous studies showing that digestates and slurries with high dry matter content sustained prolonged emissions with pronounced diurnal variability, whereas low dry matter pig slurry typically exhibits low emission rates and lacks a clear diurnal pattern (Andersson et al. 2023; Chantigny et al. 2009; Sanz-Cobena et al. 2019).

Conclusions

Although digestates exhibited higher emission than cattle and pig slurry, and emission variation within slurry types was associated with unique sources, these differences were largely accounted for by variation in pH and dry matter. This suggests that slurry type per se is not a primary driver of emission. Although rheological variables and particle size distribution may more directly reflect the physical mechanisms driving emission, their correlation with dry matter concentration and only marginal improvements in model

performance suggest limited insight or practical improvements over dry matter as a predictor. These findings support the continued measurement and use of pH and dry matter concentration as key predictors in emission models and inventories, while indicating limited added value of routine viscosity characterization under field conditions. Broader datasets across soils, climates, and application techniques are needed to reduce parameter uncertainty. Improved integration of routinely measured slurry variables into decision-support tools may enhance mitigation strategies.

Acknowledgements The authors thank the technical staff at Aarhus University (Claudia Greiner, Heidi Grønbæk, Leonid Mshanetskyi, Peter Storgård Nielsen, and Janni Ankerstjerne Sørensen) for carrying out field experiments and laboratory analysis. We would like to express our gratitude towards the biogas plants that provided digested slurry for the experiments. We would like to thank the collaborators in the MAG project for fruitful discussion of experimental work and results. Lastly, we would like to thank Thomas Kupper, Bern University of Applied Sciences and Henrik B. Møller, Aarhus University for valuable discussions about experimental design.

Author contributions J. P.: Conceptualization, Data curation, Formal analysis, Investigation, Project administration, Visualization, Writing—original draft, Writing, review & editing C. R.: Data curation, Investigation, Writing, review & editing A. S. P.: Writing, review & editing S. D. H.: Formal analysis, Methodology, Visualization, Writing – original draft, Writing, review & editing.

Funding Open access funding provided by Aarhus Universitet. Landbrugsstyrelsen, 34009-21-1829.

Data availability No datasets were generated or analysed during the current study.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Abanades S, Abbaspour H, Ahmadi A, Das B, Ehyaei MA, Esmailion F, El Haj Assad M, Hajilounezhad T, Jamali DH, Hmida A, Ozgoli HA, Safari S, AlShabi M, Bani-Hani EH (2022) A critical review of biogas production and usage with legislations framework across the globe. *Int J Environ Sci Technol (Tehran)* 19(4):3377–3400. <https://doi.org/10.1007/s13762-021-03301-6>
- Adamsen APS, and Hafner S (2021) Emissionsfaktorer for ammoniak fra ALFAM2 for ammoniakreducerende udbringningsteknikker (EN: Emission factors for ammonia from ALFAM2 for ammonia mitigating application techniques). Memorandum from DCA–National Center for Food and Agriculture, Aarhus University, delivered 17.11.2021.
- Anderson D, Burnham K (2004) Model selection and multi-model inference, vol Vol. 63, second ed. Springer-Verlag
- Andersson K, Delin S, Pedersen J, Hafner SD, Nyord T (2023) Ammonia emissions from untreated, separated and digested cattle slurry – effects of slurry type and application strategy on a Swedish clay soil. *Biosyst Eng* 226:194–208. <https://doi.org/10.1016/j.biosystemseng.2023.01.012>
- APHA (2017) Standard methods for the examination of water and wastewater, 23rd ed. American Public Health Association
- Bates D, Maechler M, Bolker B, Walker S (2015) Fitting linear mixed-effects models using lme4. *J Stat Softw* 67(1):1–48. <https://doi.org/10.18637/jss.v067.i01>
- Bittman S, van Vliet LJP, Kowalenko CG, McGinn S, Hunt DE, Bounaia F (2005) Surface-banding liquid manure over aeration slots: a new low-disturbance method for reducing ammonia emissions and improving yield of perennial grasses. *Agron J* 97(5):1304–1313. <https://doi.org/10.2134/agronj2004.0277>
- Chantigny MH, MacDonald JD, Beaupré C, Rochette P, Angers DA, Massé D, Parent L-E (2009) Ammonia volatilization following surface application of raw and treated liquid swine manure. *Nutr Cycl Agroecosyst* 85(3):275–286. <https://doi.org/10.1007/s10705-009-9266-7>
- Chuenchart W, Surendra KC, Khanal SK (2024) Understanding anaerobic co-digestion of organic wastes through meta-analysis. *ACS EST Eng*. <https://doi.org/10.1021/acses.tengg.3c00598>
- Epper CA, Zavattaro L, Velthof GL, Thuriès L, Steinsberger T, Sørensen P, Richards K, Oberson A, Möller K, Merbold L, Menzi H, Liebisch F, Levasseur F, Leiber F, Kupper T, Krol D, Janke D, Hofmeier M, Hoekstra NJ, Mayer J (2025) Optimizing nitrogen use efficiency in European livestock systems: from feed to plant growth. *Adv Agron* 191(277):362. <https://doi.org/10.1016/bs.agron.2025.01.003>
- Fangueiro D, Hjorth M, Gioelli F (2015) Acidification of animal slurry—a review. *J Environ Manage* 149:46–56. <https://doi.org/10.1016/j.jenvman.2014.10.001>
- Frost JP, Stevens RJ, Laughlin RJ (1990) Effect of separation and acidification of cattle slurry on ammonia volatilization and on the efficiency of slurry nitrogen for herbage production. *J Agric Sci* 115:49–56
- Hafner SD, Kamp JN, Pedersen J (2024) Experimental and model-based comparison of wind tunnel and inverse dispersion model measurement of ammonia emission from field-applied animal slurry. *Agric for Meteorol*. <https://doi.org/10.1016/j.agrformet.2023.109790>
- Hafner SD, Pacholski A, Bittman S, Burchill W, Bussink W, Chantigny M, Carozzi M, Génemont S, Häni C, Hansen MN, Huijsmans J, Hunt D, Kupper T, Lanigan G, Loubet B, Misselbrook T, Meisinger JJ, Neftel A, Nyord T, et al (2018) The ALFAM2 database on ammonia emission from field-applied manure: Description and illustrative analysis. *Agric For Meteorol* 258:66–79. <https://doi.org/10.1016/j.agrformet.2017.11.027>
- Hafner SD, Pedersen J, Fuß R, Kamp JN, Dalby FR, Amon B, Pacholski A, Adamsen APS, Sommer SG (2025) Improved tools for estimation of ammonia emission from field-applied animal slurry: refinement of the ALFAM2 model and database. *Atmospheric Environ*. <https://doi.org/10.1016/j.atmosenv.2024.120910>
- Häni C, Sintermann J, Kupper T, Jocher M, Neftel A (2016) Ammonia emission after slurry application to grassland in Switzerland. *Atmos Environ* 125:92–99. <https://doi.org/10.1016/j.atmosenv.2015.10.069>
- Helsel D (2010) Much ado about next to nothing: incorporating nondetects in science. *Ann Occup Hyg* 54(3):257–262. <https://doi.org/10.1093/annhyg/mep092>
- Jensen ON, Beyer M, Sorensen MK, Kreimeyer M, Nielsen NC (2021) Fast and Accurate Quantification of Nitrogen and Phosphorus Constituents in Animal Slurries Using NMR Sensor Technology. *ACS Omega* 6(27):17335–17341. <https://doi.org/10.1021/acsomega.1c01441>
- Kamp JN, Chowdhury A, Adamsen APS, Feilberg A (2019) Negligible influence of livestock contaminants and sampling system on ammonia measurements with cavity ring-down spectroscopy. *Atmos Meas Tech* 12(5):2837–2850. <https://doi.org/10.5194/amt-12-2837-2019>
- Kariyama ID, Zhai X, Wu B (2018) Physical and rheological properties of animal manure: A review. *Trans ASABE* 61(3):1113–1120. <https://doi.org/10.13031/trans.12768>
- Khoshnevisan B, Duan N, Tsapekos P, Awasthi MK, Liu Z, Mohammadi A, Angelidaki I, Tsang DCW, Zhang Z, Pan J, Ma L, Aghbashlo M, Tabatabaei M, Liu H (2021) A critical review on livestock manure biorefinery technologies: sustainability, challenges, and future perspectives. *Renew Sustain Energy Rev* 135:110033. <https://doi.org/10.1016/j.rser.2020.110033>
- Lamolinara B, Perez-Martinez A, Guardado-Yordi E, Guillen Fiallos C, Dieguez-Santana K, Ruiz-Mercado GJ (2022) Anaerobic digestate management, environmental impacts, and techno-economic challenges. *Waste Manag* 140:14–30. <https://doi.org/10.1016/j.wasman.2021.12.035>
- Langley-Randall J, Jones DL, Cotton J, Williams JR, Chadwick DR (2024) Slurry acidification is as effective as slurry injection at reducing ammonia emissions without increasing N2O emissions: a short-term mesocosm study. *Geoderma Reg*. <https://doi.org/10.1016/j.geodrs.2024.e00791>
- Liu Y, Chen J, Song J, Hai Z, Lu X, Ji X, Wang C (2019) Adjusting the rheological properties of corn-straw slurry to reduce the agitation power consumption in anaerobic digestion. *Bioresour Technol* 272:360–369. <https://doi.org/10.1016/j.biortech.2018.10.050>

- Lüdecke D, Ben-Shachar S, Patil I, Waggoner P, Makowski D (2021) Performance: an R package for assessment, comparison and testing of statistical models. *J Open Source Softw.* <https://doi.org/10.21105/joss.03139>
- McCullough MR, Pedersen J, Nyord T, Sørensen P, Melander B (2022) Ammonia emissions, exposed surface area, and crop and weed responses resulting from three post-emergence slurry application strategies in cereals. *Agronomy.* <https://doi.org/10.3390/agronomy12102441>
- Misselbrook TH, Smith KA, Johnson RA, Pain BF (2002) Slurry application techniques to reduce ammonia emissions: results of some UK field-scale experiments. *Biosyst Eng* 81(3):313–321. <https://doi.org/10.1006/bioe.2001.0017>
- Misselbrook TH, Nicholson FA, Chambers BJ (2005) Predicting ammonia losses following the application of livestock manure to land. *Bioresour Technol* 96(2):159–168. <https://doi.org/10.1016/j.biortech.2004.05.004>
- Møller HB, Sørensen P, Olesen JE, Petersen SO, Nyord T, Sommer SG (2022) Agricultural biogas production—climate and environmental impacts. *Sustainability.* <https://doi.org/10.3390/su14031849>
- Möller K, Müller T (2012) Effects of anaerobic digestion on digestate nutrient availability and crop growth: a review. *Eng Life Sci* 12(3):242–257. <https://doi.org/10.1002/elsc.201100085>
- Nakagawa S, Schielzeth H (2012) A general and simple method for obtaining R² from generalized linear mixed-effects models. *Methods Ecol Evol* 4(2):133–142. <https://doi.org/10.1111/j.2041-210x.2012.00261.x>
- Nkoa R (2013) Agricultural benefits and environmental risks of soil fertilization with anaerobic digestates: a review. *Agron Sustain Dev* 34(2):473–492. <https://doi.org/10.1007/s13593-013-0196-z>
- Nyord T, Hafner S, Adamsen APS, Sommer SG (2021) Ammoniakfordampning fra forsuret gylle ved udbringning med slæbeslange (EN: Ammonia emission from acidified slurry applied with trailing hoses) (Advisory report from DCA - Danish Centre for Food and Agriculture, Aarhus University, submitted: 15–02–2021). https://pure.au.dk/portal/files/211563336/Forsuring_150221.pdf
- Pedersen J, Hafner SD (2023) Ammonia emissions after field application of anaerobically digested animal slurry: literature review and perspectives. *Agric Ecosyst Environ.* <https://doi.org/10.1016/j.agee.2023.108697>
- Pedersen J, Andersson K, Feilberg A, Delin S, Hafner S, Nyord T (2021a) Effect of exposed surface area on ammonia emissions from untreated, separated, and digested cattle manure. *Biosyst Eng* 202:66–78. <https://doi.org/10.1016/j.biosystemseng.2020.12.005>
- Pedersen J, Nyord T, Feilberg A, Labouriau R, Hunt D, Bittman S (2021b) Effect of reduced exposed surface area and enhanced infiltration on ammonia emission from untreated and separated cattle slurry. *Biosyst Eng* 211:141–151. <https://doi.org/10.1016/j.biosystemseng.2021.09.003>
- Pedersen J, Hafner SD, Adamsen APS (2022) Effectiveness of mechanical separation for reducing ammonia loss from field-applied slurry: assessment through literature review and model calculations. *J Environ Manage* 323:116196. <https://doi.org/10.1016/j.jenvman.2022.116196>
- Pedersen J, Hafner SD, Pacholski A, Karlsson VI, Rong L, Labouriau R, Kamp JN (2024a) Evaluation of optimized flux chamber design for measurement of ammonia emission after field application of slurry with full-scale farm machinery. *Atmos Meas Tech* 17(14):4493–4505. <https://doi.org/10.5194/amt-17-4493-2024>
- Pedersen J, Hafner SD, Romio C, Pacholski AS (2026) Assessment of ammonia emission mitigation efficiency of established and novel field application techniques for anaerobically digested slurry. *Atm Environ X*:29. <https://doi.org/10.1016/j.aeaoa.2026.100415>
- Pedersen J, Labouriau R, Feilberg A (2024b) Effect of slurry separation and air-plasma treatment on NH₃ and VOC emissions from field applied biogas digestate and pig slurry to grassland. *Biosyst Eng* 247:257–266. <https://doi.org/10.1016/j.biosystemseng.2024.09.014>
- Romio C, Ward AJ, Møller HB (2024) Characterization and valorization of biogas digestate and derived organic fertilizer products from separation processes. *Front Sustain Food Syst* 8:1415508. <https://doi.org/10.3389/fsufs.2024.1415508>
- Salazar F, Martínez-Lagos J, Alfaro M, Misselbrook T (2014) Ammonia emission from a permanent grassland on volcanic soil after the treatment with dairy slurry and urea. *Atmos Environ* 95:591–597. <https://doi.org/10.1016/j.atmosenv.2014.06.057>
- Sanz-Cobena A, Misselbrook TH, Hernáiz P, Vallejo A (2019) Impact of rainfall to the effectiveness of pig slurry shallow injection method for NH₃ mitigation in a Mediterranean soil. *Atmos Environ* 216:116913. <https://doi.org/10.1016/j.atmosenv.2019.116913>
- Schneider N, Gerber M (2020) Rheological properties of digestate from agricultural biogas plants: an overview of measurement techniques and influencing factors. *Renew Sustain Energy Rev.* <https://doi.org/10.1016/j.rser.2020.109709>
- Simmons GF (1996) Calculus with analytic geometry. McGraw-Hill Education
- Smith KA, Jackson DR, Misselbrook TH, Pain BF, Johnson RA (2000) PA—precision agriculture. *J Agric Eng Res* 77(3):277–287. <https://doi.org/10.1006/jaer.2000.0604>
- Sommer SG, Husted S (1995a) The chemical buffer system in raw and digested animal slurry. *J Agric Sci* 124:45–53
- Sommer SG, Husted S (1995b) A simple model of pH in slurry. *J Agric Sci* 124:447–453
- Sommer SG, Misselbrook TH (2015) A review of ammonia emission measured using wind tunnels compared with micrometeorological techniques. *Soil Use Manag* 32(S1):101–108. <https://doi.org/10.1111/sum.12209>
- Sommer SG, Jensen LS, Clausen SB, SØgaard HT (2006) Ammonia volatilization from surface-applied livestock slurry as affected by slurry composition and slurry infiltration depth. *J Agric Sci* 144(3):229–235. <https://doi.org/10.1017/s0021859606006022>
- Team, R. C. (2025) R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. In <https://www.R-project.org/>
- Thompson RB, Pain BF, Lockyer DR (1990) Ammonia volatilization from cattle slurry following surface application to grassland. I. Influence of mechanical separation, changes

- in chemical composition during volatilization and the presence of the grass sward. *Plant Soil* 125:109–117
- van der Weerden TJ, Noble A, de Klein CAM, Hutchings N, Thorman RE, Alfaro MA, Amon B, Beltran I, Grace P, Hassouna M, Krol DJ, Leytem AB, Salazar F, Velthof GL (2021) Ammonia and nitrous oxide emission factors for excreta deposited by livestock and land-applied manure. *J Environ Qual* 50(5):1005–1023. <https://doi.org/10.1002/jeq2.20259>
- Webb J, Pain B, Bittman S, Morgan J (2010) The impacts of manure application methods on emissions of ammonia, nitrous oxide and on crop response—a review. *Agric Ecosyst Environ* 137(1–2):39–46. <https://doi.org/10.1016/j.agee.2010.01.001>
- Weiland P (2009) Biogas production: current state and perspectives. *Appl Microbiol Biotechnol* 85(4):849–860. <https://doi.org/10.1007/s00253-009-2246-7>
- Zilio M, Pigoli A, Rizzi B, Geromel G, Meers E, Schoumans O, Giordano A, Adani F (2021) Measuring ammonia and odours emissions during full field digestate use in agriculture. *Sci Total Environ* 782:146882. <https://doi.org/10.1016/j.scitotenv.2021.146882>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.